EE 301
Signals and Systems

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Outline

1. Transformations of the independent variable
2. Important properties of signals
   - Periodic signals
   - Even and Odd Signals
3. Basic CT & DT signals
   - CT Complex Exponential & Sinusoidal Signals
   - DT Complex Exponential & Sinusoidal Signals
   - DT Unit Impulse & Unit Step Sequences
   - CT Unit Impulse & Unit Step Functions
4. Systems
5. Basic System Properties
   - Memory, Causality, Invertibility, & Stability
   - Time Invariance
   - Linearity

Transformations of the independent variable

We sometimes work with signals after modifying the independent variable (i.e. time axis).

Example

Fast forward

- **Time Shift** : $x(t - t_0)$
- **Time Reversal** : $x(-t)$
- **Time Scaling** : $x(\frac{t}{a})$ or $x(at)$, a: real number

Note that $x(at)$ is always defined for $a \epsilon \mathbb{R}$. The same is not true for $x[an]$.

Example

$x[n]$: a discrete sequence. Which of the following are defined: $x[2n]$, $x[\sqrt{2}n]$, $x[\frac{n}{2}]$?

Example

Find $y(t) = x(-t+1)$ for the given $x(t)$.

Periodic signals

Definition

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A CT signal $x(t)$ is said to be periodic if there is a value T for which

$$x(t) = x(t + T) \text{ holds for all } t$$

In this case, we say that $x(t)$ is periodic with T.

Periodicity is defined similarly for DT signals: periodic with N if $x[n] = x[n + N]$ for all n where N is an integer.

Example

Examples of periodic CT and DT signals

Note that if $x(t)$ is periodic with T then it is also periodic with $2T$, $3T$, $4T$,....

- **Fundamental period** T_0 of $x(t)$ (N_0 of $x[n]$): the smallest positive T (N) for which the above equalities hold.

Example (Challenge yourself!)

Prove whether $x_p(t) = \sum_{m=-\infty}^{\infty} x(t - mT)$ is periodic or not, when T: some constant, $x(t)$: arbitrary CT signal.

Example (Challenge yourself!)

If $x(t)$ is periodic with T, how about $x(t + t_0)$ and $x(\frac{t}{a})$? Similarly, if $x[n]$ is periodic with N, how about $x[n + n_0]$ and $x[Mn]$?

Even and Odd Signals

A signal is even if

Even and Odd Signals

A signal is even if $x(-t) = x(t)$ for all t. (DT: $x[-n] = x[n]$)

A signal is odd if

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A signal is even if $x(-t) = x(t)$ for all t. (DT: $x[-n] = x[n]$)

A signal is odd if $x(-t) = -x(t)$ for all t. $(x[-n] = -x[n])$

Example

Examples of even and odd signals

Example (Challenge yourself!)

Can we decompose arbitrary signals into a sum of two signals, one of which is even and the other is odd:

$$x(t) = x_{ev}(t) + x_{od}(t)$$

- Is the decomposition unique?
- Closed-form expression for the decomposition?

Check the book for the solution:

$$x_{ev}(t) = \frac{1}{2}(x(t) + x(-t)), \quad x_{od}(t) = \frac{1}{2}(x(t) - x(-t))$$

These are called even and odd parts of $x(t)$.

CT Complex Exponential & Sinusoidal Signals

[Short digression: Review of complex numbers (Page 71)]

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The CT complex exponential signal is generally of the form

$$x(t) = C\, e^{at}$$

where C and a are in general complex numbers.

1. **Real Exponential Signals:**
   C and a are real numbers
   - If $a > 0$, growing exponential:
   - If $a < 0$, decaying exponential:

2. Periodic Complex Exponential Signals:
a is purely imaginary

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$$x(t) = Ce^{j\omega_0 t} \qquad (\omega_0\text{: radians/second})$$

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$\Rightarrow$ Verify periodicity:

- Fundamental period $T_0 = \frac{2\pi}{|\omega_0|}$
- $e^{j\omega_0 t}$ and $e^{-j\omega_0 t}$ have the same fundamental period.
- Sets of harmonically related periodic exponentials:
- Euler's relations:
- Does $x(t)$ have finite average power? finite energy?

3. Sinusoidal Signals:

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- odd signal when $\theta = \frac{\pi}{2} + k\pi$

Example (Challenge yourself!)

How to express $x(t)$ in terms of periodic complex exponentials?

4. **General Complex Exponential Signals:**
Both C and a are complex.

Let us represent C in polar form as $C = |C|e^{j\theta}$ and a in rectangular form as $a = \alpha + j\omega_0$. Then
$x(t) = Ce^{at} =$

- $\alpha = 0$: both real and imaginary parts are
- $\alpha > 0$: both real and imaginary parts are
- $\alpha < 0$: both real and imaginary parts are

Example

$x(t) = 2e^{(3+j\omega_0)t}$
$x(t) = e^{(-1+j\omega_0)t}$

DT Complex Exponential & Sinusoidal Signals

The DT complex exponential signal is generally of the form

$$x[n] = C\alpha^n \quad \text{or} \quad x[n] = Ce^{\beta n} \; (\alpha = e^{\beta})$$

where C and α are in general complex numbers.

1. **Real Exponential Signals:**
   C and α are reals.

Example

DT real exponential signals with various behaviors

2. **Sinusoidal Signals:**
   If β is purely imaginary (i.e. $|\alpha| = 1$), we obtain $x[n] = Ce^{j\Omega_0 n}$. This is related to sinusoidal signals:

3. **General Complex Exponentials (Damped sinusoids):**
   β is not purely imaginary.

   By representing C and α in polar form,

$$\begin{aligned} x[n] &= C\alpha^n = |C|e^{j\theta}(|\alpha|e^{j\Omega_0})^n = |C||\alpha|^n e^{j(\Omega_0 n+\theta)} \\ &= |C||\alpha|^n(\cos(\Omega_0 n+\theta) + j\sin(\Omega_0 n+\theta)) \end{aligned}$$

- $|\alpha| > 1$: both real and imaginary parts are ...
- $|\alpha| < 1$: both real and imaginary parts are ...

Periodicity Properties of DT Complex Exponentials

Although there are many similarities between CT and DT signals, there are also some important differences. One such difference exists between $e^{j\omega_0 t}$ and $e^{j\Omega_0 n}$.

Remember:

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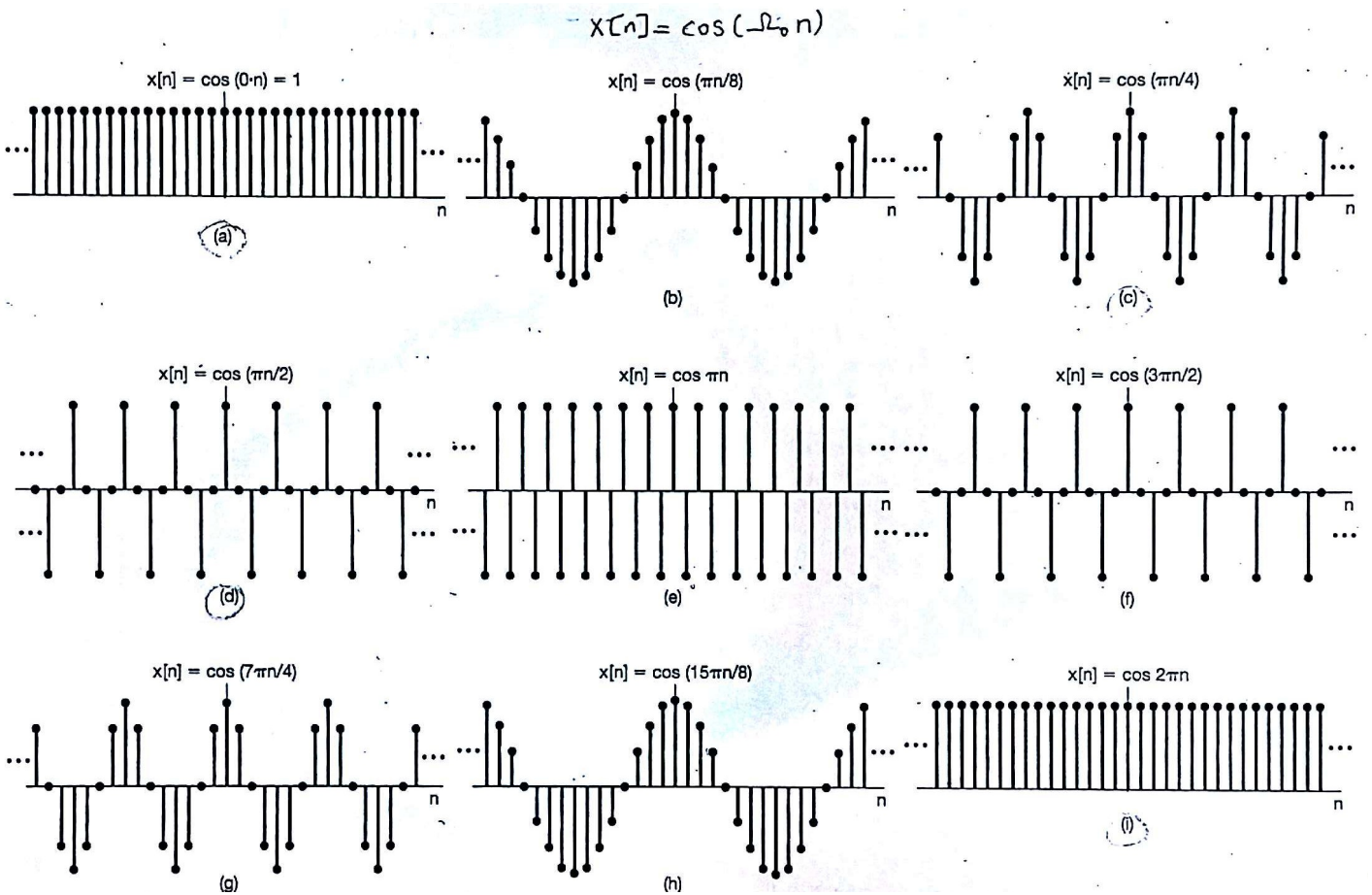
Both of the above properties are different for $e^{j\Omega_0 n}$.

DT sinusoidal sequences for several frequencies



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For $e^{j\Omega_0 n}$:

- As $|\Omega_0|$ increases, the rate of oscillation does not increase continuously.
- $e^{j\Omega_0 n}$ is not even distinct for different Ω_0; identical for $\Omega_0 + 2\pi k$
- Periodicity? $e^{j\Omega_0 n}$ is periodic only if Ω_0 can be written in the form $\Omega_0 = 2\pi \frac{k}{N}$ for some integers $N > 0$ and k.

Example

Is $x[n] = e^{jn}$ periodic? If so, find the fundamental period and frequency.

Example

Is $x[n] = 5\cos(\pi n)$ periodic? If so, find the fundamental period and frequency.

DT Unit Impulse & Unit Step Sequences

- DT unit impulse is defined as

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0. \end{cases}$$

- DT unit step is defined as

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0. \end{cases}$$

- The relation between $u[n]$ and $\delta[n]$:
  $\delta[n] =$
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  $\delta[n] =$
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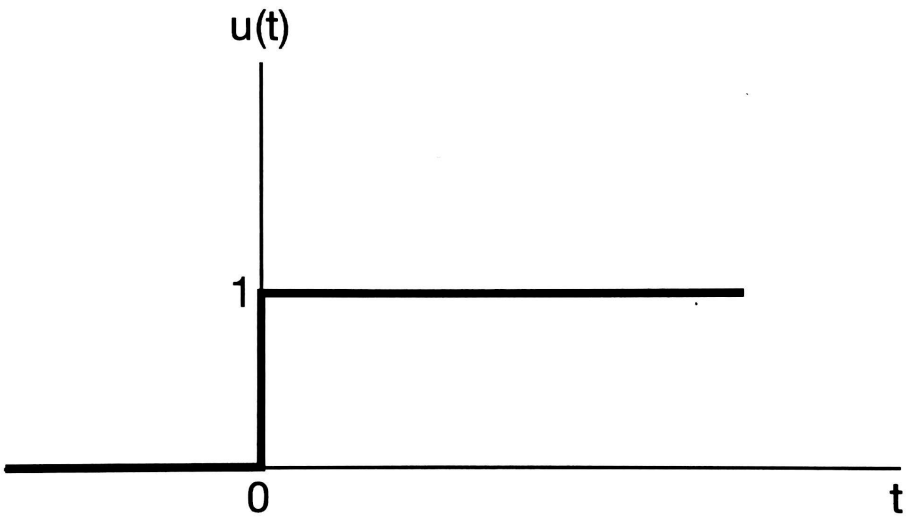
(Sum of Shifted Impulses / Running Sum Interpretations)

Properties of DT Unit Impulse $\delta[n]$

- $$\sum_{n=-\infty}^{\infty} \delta[n] = \qquad , \qquad \sum_{n=n_1}^{n=n_2} \delta[n] =$$
- $x[n]\delta[n] =$
  $(x[n]\delta[n-n_0] = \qquad\qquad\qquad\qquad)$ (Sampling Prop.)
- $\sum_{n=-\infty}^{\infty} x[n]\delta[n] =$
  $\sum_{n=n_1}^{n=n_2} x[n]\delta[n] =$
  $(\sum_{n=-\infty}^{\infty} x[n]\delta[n_0-n] = \qquad\qquad\qquad)$ (Convolution Prop.)

CT Unit Step & Unit Impulse Functions

The CT unit step function: $u(t) = \begin{cases} 1, & t > 0 \\ 0, & t < 0. \end{cases}$



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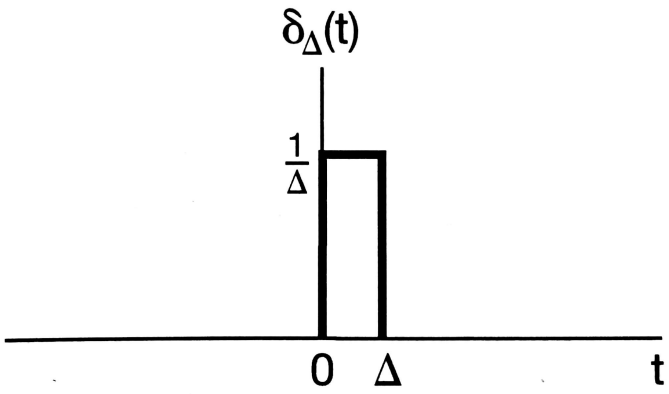
however, this is problematic in the classical sense since $u(t)$ is formally not differentiable at $t = 0$ ("generalized derivative").

For a more formal development, let us define new functions (approximations to the original step & impulse):

That is, more formally, the unit impulse func. is defined as

$$\delta(t) = \lim_{\Delta \to 0} \delta_\Delta(t)$$



- Note that $\delta_\Delta(t)$ is a short pulse of duration Δ and with unit area for any value of Δ. $\delta(t)$ should be viewed as an idealization of the short pulse when the duration Δ is very small (for any practical purpose).
- In practical terms, you can think of $\delta(t)$ as any func. of unit area, concentrated very near $t = 0$.

Properties of unit impulse function $\delta(t)$

- It is a signal of unit area vanishing everywhere except at origin.
- It is the (generalized) derivative of the unit step func.
- For any cont. func. $x(t)$,
  - $x(t)\delta(t) =$
    $x(t)\delta(t - t_0) =$
  - $\int_{-\infty}^{\infty} x(t)\delta(t)dt =$
    $\int_{t_1}^{t_2} x(t)\delta(t)dt =$
    $(\int_{-\infty}^{\infty} x(t)\delta(t_0 - t)dt = \quad)$

Example (Challenge yourself!)

Prove the following:
Running Integral Interpretation : $u(t) = \int_{-\infty}^{t} \delta(\tau) d\tau$

Example

Prove the following:
Moving Impulse Interpretation : $u(t) = \int_{0}^{\infty} \delta(t - \tau) d\tau$

Example (Challenge yourself!)

Example 1.7 in the book (square type signal)

Systems

Definition

System: Any process that results in the transformation of signals.

Example

Examples of systems

CT and DT systems:

Interconnection of systems:

Memory

Definition

A system is memoryless (instantaneous) if its output at any time instant depends only on the value of the input at that particular instant.

Example

Examples

Causality

Definition

A system is causal if its output at any time depends only on the values of the input at present time and/or in the past.

Example

Examples

- Physical systems that process time-domain signals in real-time

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- Physical systems that process time-domain signals in real-time can not respond before they are stimulated. In this case, causality is equivalent to realizability.
- Non-causal systems are of interest when the independent variable is not time or offline processing is possible.

Example (Challenge yourself!)

Causal or not:

- $y[n] = x[3n]$
- $y[n] = \sum_{m=-\infty}^{n} \alpha_m x[n-m]$
- $y(t) = x(t)^2$
- $y(t) = \int_{-\infty}^{t} x(\tau) d\tau$
- a memoryless system

Invertibility

Definition

A system is invertible if for distinct inputs, $x_1(t) \neq x_2(t)$, it generates distinct outputs, $y_1(t) \neq y_2(t)$.

If a system is invertible, then a corresponding inverse system exists.

Example

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Example (Challenge yourself!)

Invertible or not: $y(t) = 5x(t - 1/2)$?

Stability

Definition (Bounded Input Bounded Output (BIBO) stab.)

A system is stable if bounded inputs lead to bounded outputs.

Example

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Stability

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Example (Challenge yourself!)

Stable or not:

- $y(t) = \frac{x(t)}{x(t-1)+1}$
- $y(t) = e^{\alpha t}x(t)$, $\alpha \in \mathcal{C}$
- $y(t) = \frac{d}{dt}x(t)$

Time Invariance

Definition

A system is time-invariant if a time shift in the input signal causes same amount of time shift in the output signal.

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Example (Challenge yourself!)

Time-invariant or not:

- $y(t) = \sin(x(t))$
- $y(t) = x(2t)$
- $y(t) = e^t\, x(t)$
- $y[n] = n\, x[n]$

Linearity

Definition

A linear system has the superposition property:

Example

Examples

Example (Challenge yourself!)

Linear or not:

- $y[n] = 5x[n] + 2$
- $y(t) = x^2(t)$
- $y(t) = \frac{d^n x(t)}{dt^n}$